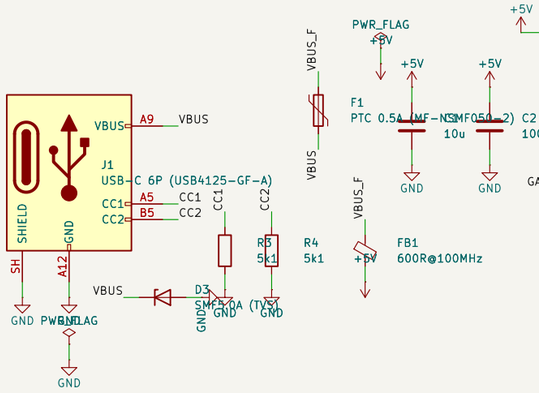
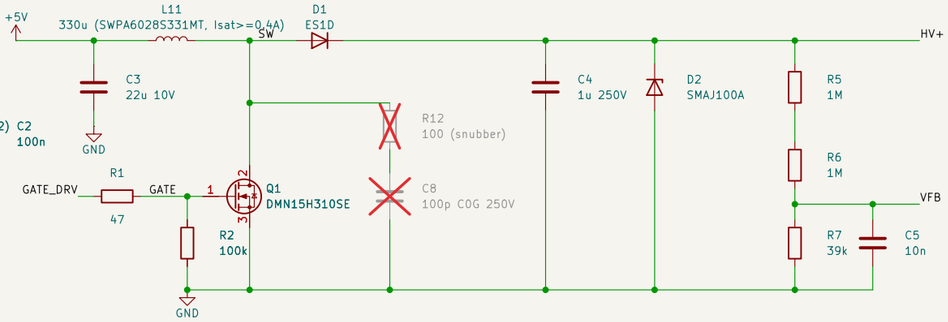
POWER IN - USB-C 5V, TVS D3, PTC F1, ferrite FB1



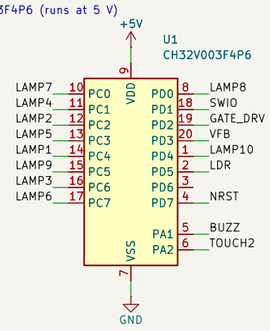
HV BOOST 75-100 V (MCU-controlled, TIM1 30 kHz)

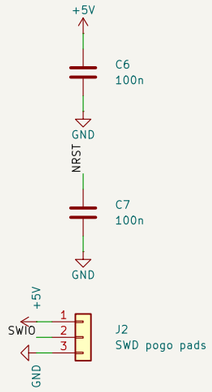


Q1 DMN15H310SE: 150 V, 0.33 Ω @ Vgs=5 V. R2 keeps boost off during reset.

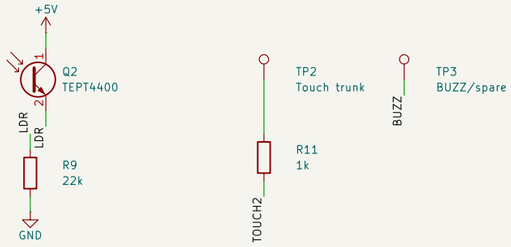
D2 = hard OVP clamp (Vbr 111-123 V). R12/C8 snubber: optional (DNP).

MCU CH32V003F4P6 (runs at 5 V)

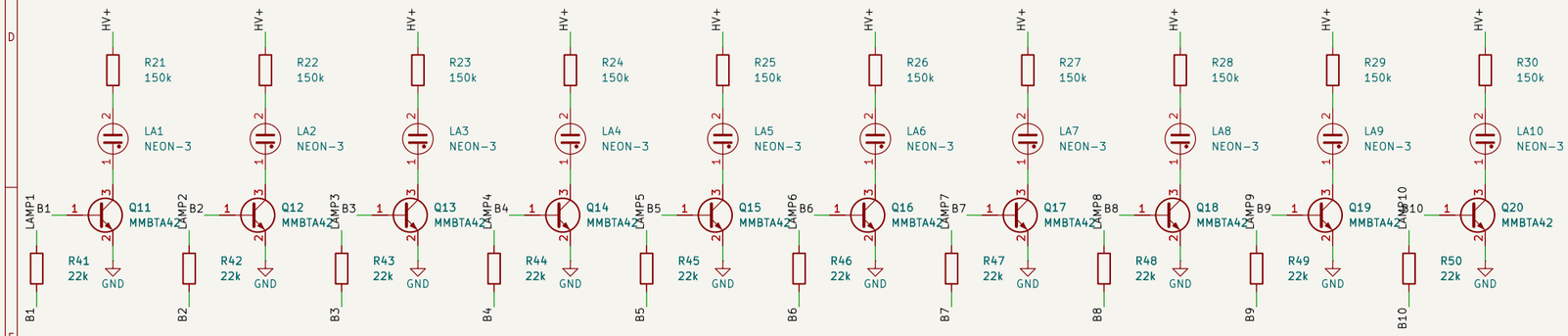




INPUTS - LDR, TKEY touch pads, spare pad



NEON LAMPS - Brightmaster NEON-3 (4x10, strike ≤90 VDC, ~60 V settled): I = (V_HV − 60 V)/150k ≈ 0.1-0.3 mA, MMBTA42 low-side switches



Sheet: /		
File: NeonPine.kicad_sch		
Title: NeonPine - neon Christmas tree night light		
Size: A3	Date: 2026-10-05	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 1/1